



AKD AGENT DESIGN PROCESS

CARE: Collaborative Agent Reasoning Engineering Agent Design Process



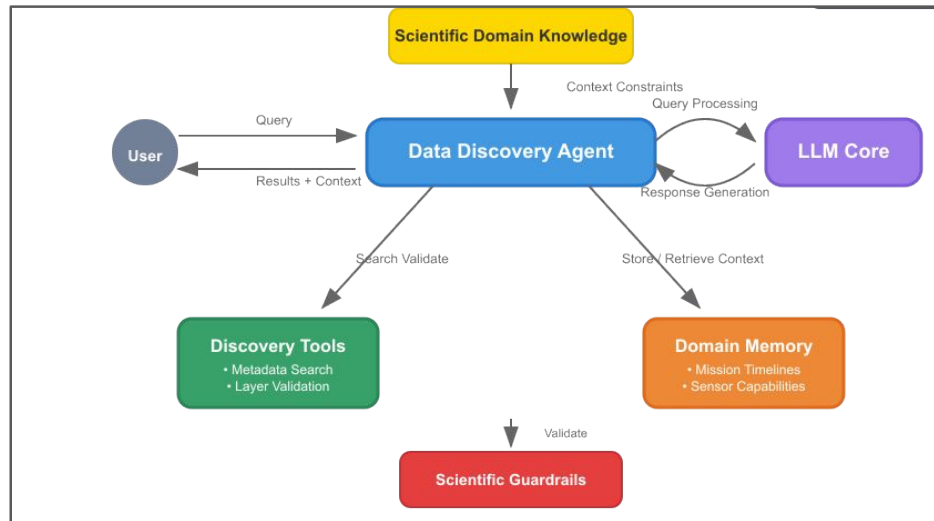
NASA Technical Memo on CARE

Date: May 19th 2026

2nd ESA-NASA Workshop on AI FMs for EO

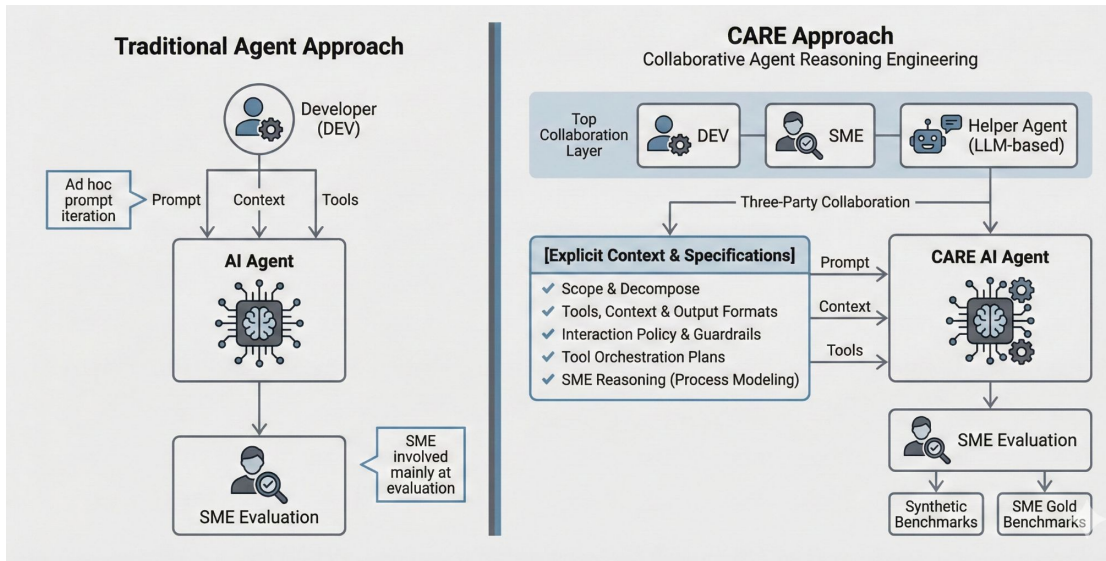


What are AI AGENTS



A specialized AI agent is an LLM connected to specific tools, has context and are designed to perform a specialized task.

Building AI AGENTS



- Operational constraints
- Edge cases
- Failure modes
- Output expectations

• **Why traditional Agent doesn't work for Agent in Science:** Move from ad-hoc prompting to disciplined agent engineering.

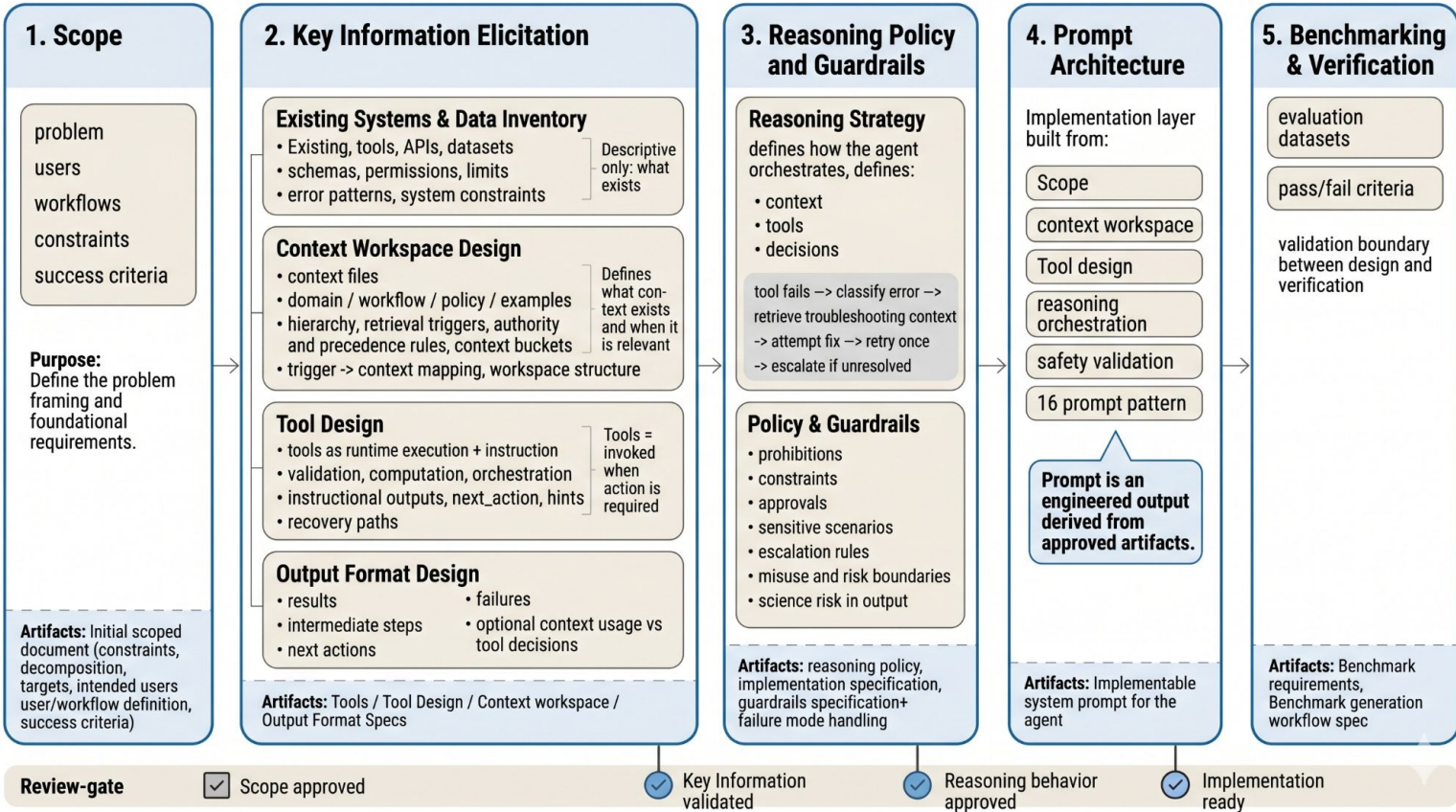
• Focus on behavior building, not just feature building

• Three-party collaboration: SMEs, Developers, Helper Agents

• Goal: Make expert reasoning explicit, reviewable, and verifiable.

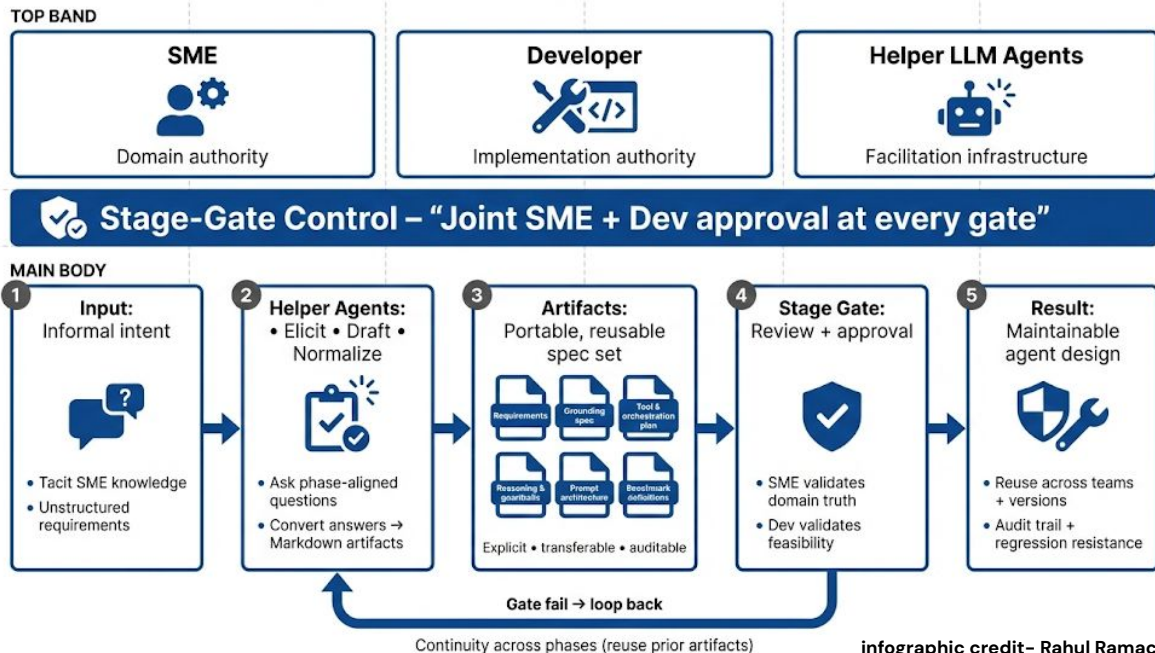


CARE* AI Agent Design Process





Three Party Collaboration



- Helper agents translate informal intent into structured artifacts

- Artifacts reviewed and approved at explicit stage gates

infographic credit– Rahul Ramachandran

[LINK to the Technical Memo](#)

Example of CARE designed Agent

Case study: Earth Science Code Search Agent

- CARE-designed agent outperforms baseline with same model & tools
- Higher Recall@1, @3, @5 on Gold Standard Benchmarks.

